

DATA 615 - Regression: Homework - 12

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1 Problem 14.13 (p. 627. 15 pts)

- In part (a), please find examples of “fitted response function” from textbook Equations (14.28), (14.30), (14.35) on p. 565, 566, 570, respectively.
- In part (c), plug the x-values ($x_1 = 50$, $x_2 = 3$) into the fitted response function in part (a). Then you can find the estimated probability.

Car purchase. A marketing research firm was engaged by an automobile manufacturer to conduct a pilot study to examine the feasibility of using logistic regression for ascertaining the likelihood that a family will purchase a new car during the next year. A random sample of 33 suburban families was selected. Data on annual family income (X1, in thousand dollars) and the current age of the oldest family automobile (X2, in years) were obtained. A follow up interview conducted 12 months later was used to determine whether the family actually purchased a new car ($Y = 1$) or did not purchase a new car ($Y = 0$) during the year.

Multiple logistic regression model (14.41) with two predictor variables in first-order terms is assumed to be appropriate.

```
cp <- read.table("./data/CH14PR13.txt", header = F)
colnames(cp) <- c("Y_new_car", "X1_fam_income", "X2_auto_age")
glimpse(cp)
```

```
## Rows: 33
## Columns: 3
## $ Y_new_car      <dbl> 0, 0, 1, 0, 0, 1, 1, 1, 0, 1, 1, 0, 0, 1, 0, 0, 1, 0, 1, ~
## $ X1_fam_income  <dbl> 32, 45, 60, 53, 25, 68, 82, 38, 67, 92, 72, 21, 26, 40, ~
## $ X2_auto_age    <dbl> 3, 2, 2, 1, 4, 1, 2, 5, 2, 2, 3, 5, 3, 4, 3, 1, 2, 3, 4, ~
```

```
summary(cp)
```

```
##      Y_new_car      X1_fam_income      X2_auto_age
## Min.   :0.0000    Min.   :10.00    Min.   :1
## 1st Qu.:0.0000    1st Qu.:22.00    1st Qu.:2
## Median :0.0000    Median :32.00    Median :3
## Mean   :0.4242    Mean   :38.24    Mean   :3
## 3rd Qu.:1.0000    3rd Qu.:53.00    3rd Qu.:4
## Max.   :1.0000    Max.   :92.00    Max.   :6
```

a. Find the maximum likelihood estimates of β_0 , β_1 , and β_2 . State the fitted response function.

```
cp_logit <- glm(Y_new_car ~ X1_fam_income + X2_auto_age, family = binomial(link = logit), data = cp)
summary(cp_logit)
```

```
##
## Call:
## glm(formula = Y_new_car ~ X1_fam_income + X2_auto_age, family = binomial(link = logit),
##      data = cp)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -4.73931    2.10195  -2.255   0.0242 *
## X1_fam_income  0.06773    0.02806   2.414   0.0158 *
## X2_auto_age    0.59863    0.39007   1.535   0.1249
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##    Null deviance: 44.987  on 32  degrees of freedom
## Residual deviance: 36.690  on 30  degrees of freedom
## AIC: 42.69
##
## Number of Fisher Scoring iterations: 4
```

The maximum likelihood estimates for the coefficients are:

- $\hat{\beta}_0 = -4.73931$
- $\hat{\beta}_1 = 0.06773$
- $\hat{\beta}_2 = 0.59863$

The fitted response function is:

$$\hat{\pi} = \frac{\exp(-4.73931 + 0.06773(\text{income}) + 0.59863(\text{age}))}{1 + \exp(-4.73931 + 0.06773(\text{income}) + 0.59863(\text{age}))}$$

b. Obtain $\exp(b_1)$ and $\exp(b_2)$ and interpret these numbers.

```
exp(cp_logit$coefficients) # Odds Ratio
```

```
##      (Intercept) X1_fam_income    X2_auto_age  
##      0.008744682    1.070079093    1.819627221
```

- $\exp(\hat{\beta}_1) = 1.07$, meaning that, after controlling for Age of the car, for every 1000 increase in family income, the odds ratio of buying a new car is 1.07.
- $\exp(\hat{\beta}_2) = 1.82$, meaning that, after controlling for family income in thousands of dollars, for each year increase in current age of the car, the odds ratio of buying a new car is 1.82.

c. What is the estimated probability that a family with annual income of \$50 thousand and an oldest car of 3 years will purchase a new car next year?

```
predict(cp_logit,  
        newdata = data.frame(X1_fam_income = 50,  
                              X2_auto_age = 3),  
        type = "response")
```

```
##           1  
## 0.6090245
```

- The estimated probability that a family, with annual income of 50 thousand dollars and an oldest car of 3 years, will buy a car in the next year is 0.609.

2 Problem 14.19 (a, b) (p. 629, 10 pts + 10 extra)

- In part (a), use Bonferroni adjustment for the joint confidence interval.
- In part (b), the “Wald test” is to use the Standard Normal (Z) distribution for the test.
- You can work on part (c) and (d) for extra credit. (extra 10 pts)

Refer to Car purchase Problem 14.13. Assume that the fitted model is appropriate and that large-sample inferences are applicable.

a. Obtain joint confidence intervals for the family income odds ratio $\exp(20\beta_1)$ for families whose incomes differ by 20 thousand dollars and for the age of the oldest family automobile odds ratio $\exp(2\beta_2)$ for families whose oldest automobiles differ in age by 2 years, with family confidence coefficient of approximately 0.90. Interpret your intervals.

```
g <- 2
fam_alpha <- 0.10
ci_beta <- confint(cp_logit, level = 1 - (fam_alpha / g))
```

```
## Waiting for profiling to be done...
```

```
exp(20 * ci_beta[2, ]) # 95% CI for exp(20/B1)
```

```
##      2.5 %      97.5 %
## 1.477058 13.950959
```

```
exp(2 * ci_beta[3, ]) # 95% CI for exp(2/B2)
```

```
##      2.5 %      97.5 %
## 0.7832232 18.0629946
```

- The family Confidence Interval for the odds ratios $\exp(20\beta_1)$ and $\exp(2\beta_2)$ are: (1.477058, 13.950959) and (0.7832232, 18.0629946), respectively, at 90% confidence level.

b. Use the Wald test to determine whether X_2 , age of oldest family automobile, can be dropped from the regression model; use $\alpha = 0.05$. State the alternatives, decision rule, and conclusion. What is the approximate P-value of the test?

```
summary(cp_logit)
```

```
##
## Call:
## glm(formula = Y_new_car ~ X1_fam_income + X2_auto_age, family = binomial(link = logit),
##      data = cp)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -4.73931    2.10195  -2.255   0.0242 *
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## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
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## AIC: 42.69
##
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```

- Hypothesis:
 - $H_0 : \beta_2 = 0$, meaning that X_2 can be dropped from the model.
 - $H_a : \beta_2 \neq 0$, meaning that X_2 should be in the model.
- The decision rule:
 - If p-value $>$ alpha or $|z\text{-statistic}| <$ z-critic: do not reject H_0 .
 - If p-value $<$ alpha or $|z\text{-statistic}| >$ z-critic: reject H_0 .
- The z-statistic (or the Wald test statistic), is computed in the regression output provided above. For X_2 , the z-statistic = 1.535 and p-value = 0.1245. For $\alpha = 0.05$, the z-critic = 1.96. Therefore, since the z-statistic = 1.532 is smaller than 1.96, with a p-value = 0.1245 larger than $\alpha = 0.05$, we do not reject H_0 . Therefore, there is sufficient evidence that X_2 can be dropped from the model, at 95% confidence level.

This is the end of Homework 12.